

Response to the Editor and Prior Reviewers

Prior manuscript ID: TNSM-2026-10776

Prior title: *Energy-Aware Intrusion Detection System with Q-learning-Based Dynamic Scheduler on Lightweight IoT Devices*

Current title: *When Learned IDS Model Selection Collapses to a Static Choice: A Measurement Study at the Edge*

Nuonan Ouyang, Adrian Shatte, Zhigang Lu, Chao Chen, and Wei Xiang
10 September 2026

Revision summary

We thank the editor and reviewers for identifying weaknesses in methodological novelty, RL formulation, baseline strength, statistics, scalability, reproducibility, and experimental rigor. We did not restore or tune the prior numerical claims. Instead, we rebuilt the dataset lineage, detector pipeline, scheduler implementation, and physical evidence, and we now retain results even when they weaken the original claim.

The primary experiment contains ten trace-matched physical blocks on a Raspberry Pi 4B 8 GB, with Static-MedRF, CFSSM, Tabular-Q, and DQN in each block. Relative to Static-MedRF, Tabular-Q saves 13.068 J per approximately 500-second run (95% paired CI 9.736–16.400 J) and DQN saves 6.227 J (1.600–10.853 J), but both reduce F1 by 1.343 percentage points and recall by 2.954 percentage points and select LightLR in every formal window. We therefore no longer claim that the saving comes from dynamic switching.

We then added strengthening controls without changing the primary experiment. The matched Static-LightLR control shows Tabular-Q minus Static-LightLR energy of +0.455 J with 95% CI $[-1.861, 2.772]$ J and identical detection; The scheduler-overhead and policy-sensitivity analyses isolate timing and robustness. Two later reviewer controls were frozen before their own execution. A fixed-rate confidence-cascade campaign contains 40 physical runs. A separate variable-load campaign measures six methods in 60 runs over a fixed 3×3 load/prevalence factorial. Under variable load, DeadlineGuard selects MedRF in 3,500 and LightLR in 1,500 windows, uses 43.714 J less than Static-MedRF (95% paired CI 41.418–46.010 J; 2.98%), and eliminates Static-MedRF’s 512/5,000 deadline misses at a 1.373-pp F1 cost. The frozen DQN remains LightLR, whereas frozen Tabular-Q reaches unseen tabular rows whose zero-valued ties resolve to the first configured action, TinyDT, in all 5,000 windows and produces FPR 67.76%. The planned controller-only power arm stops before measurement because its predeclared four-state subset is not uniquely identifiable; The TinyDT diagnostic retains threshold 0.5 while reporting a validation-selected TinyDT calibration diagnostic. The final paper therefore distinguishes useful conditional control from unsupported learned adaptation.

Associate Editor

Prior reviewer comment

The manuscript addresses a relevant IoT IDS scheduling problem with practical implementation. However, significant concerns remain regarding methodological novelty, RL formulation, baseline comparisons, statistical validation, scalability, reproducibility, and experimental rigor. Substantial technical revision and stronger evaluation are required before the work can be reconsidered.

Author response

We agree. Tabular Q-learning is not presented as a methodological advance. The paper instead asks which detector schedule a controller actually executes and whether the resulting whole-device operating point survives comparison with static and simple conditional controls. In addition to DQN, causal state/action/reward definitions, the matched-control, overhead, and policy-sensitivity studies, and the independent archive, we added the fixed-rate confidence-cascade and variable-load/DeadlineGuard physical campaigns. Where the variable-load campaign shows the frozen learned policies fail to realize useful adaptation, we retain that outcome. The controller-only arm’s unresolved predeclared state subset is reported as a stop rather than being selected after the fact.

Table 1: *
Evidence status in the rebuilt study.

Concern	Final rebuilt evidence	Status / boundary
RL formulation	Explicit five-field causal state, 324-state encoding, reward normalization, frozen Q/DQN configurations	Closed descriptively
Relevant static control	Ten matched matched-control blocks with Static-LightLR, Q, DQN, Static-MedRF	Closed as post-hoc attribution evidence
Physical statistics	Ten paired blocks, Student- <i>t</i> CIs, exact sign-flip robustness, raw energy reconstruction	Closed for declared effects
Decision overhead	Corrected 240-unit scheduler-overhead replacement with raw timing arrays and telemetry	Closed for state-to-action timing
Policy variability	10 new Q seeds + 10 new DQN seeds	Closed on held-out support
Sensitivity	85 the reward/state sensitivity analysis + 35 the hyperparameter sensitivity analysis valid policies	Closed as post-hoc sensitivity
Fixed confidence cascade	Fixed-rate cascade: 40 runs, 10 paired blocks, band [0.3, 0.7]	Closed prospectively for fixed-rate cascade campaign
Variable load and deadlines	Variable load: 60 runs, 10 paired blocks, fixed 3×3 factorial	Closed prospectively for variable-load campaign
Controller-only power	Controller-only power arm stopped before measurement: four-state subset ambiguous	Explicit NOT_EXECUTED boundary
TinyDT calibration	TinyDT threshold diagnostic validation-selected threshold applied once to test	Post-hoc diagnostic; primary unchanged
Executed-source provenance	6/6 historical source bytes recovered and hash-matched	Closed
Scalability/generalization	One primary Pi; TON-only dynamic energy	Explicitly scoped, not claimed

Location / evidence

Introduction; Sections on frozen detectors, evaluation protocol, physical scheduler results, Discussion, Limitations; Supplementary Tables S14-S18.

Reviewer 1

Prior reviewer comment

The manuscript tackles a relevant IoT scheduling problem with a practical prototype; however, the current contribution is largely engineering-oriented, with limited methodological novelty and insufficient validation against state-of-the-art approaches.

Author response

We agree that the rebuilt paper should not rely on engineering novelty alone. We therefore make the system contribution falsifiable through stronger comparators and evidence: the same four-detector pool is evaluated with a rule controller, Tabular-Q, and DQN; physical whole-device effects use matched blocks; matched-control adds the static detector actually realized by the learned policies; scheduler-overhead isolates state-to-action overhead; and multi-seed and sensitivity analyses test whether the realized policy persists across new training seeds and predeclared reward/state/hyperparameter perturbations. The negative result is retained: under the observed support, both learned controllers collapse to a static LightLR operating point. The claimed contribution is consequently measurement-grounded service-management evidence and evaluation methodology, not a new RL algorithm.

Location / evidence

Introduction; Evaluation Protocol; the matched-control, overhead, and policy-sensitivity studies results; Discussion and Limitations.

Prior reviewer comment

Comment 1: The use of tabular Q-learning is too basic for this problem setting. More advanced approaches (e.g., DQN, PPO) are widely used for dynamic scheduling and would provide stronger generalization in larger state spaces.

Author response

We added DQN under the same 324-state/four-action interface. The result is informative precisely because greater function-approximation complexity does not change the realized detector schedule: the primary Q and DQN policies both select LightLR in all formal windows, and the multi-seed analysis shows the same held-out LightLR behavior for all ten new seeds per algorithm. matched-control further shows that both learned methods reproduce Static-LightLR detection exactly. The corrected scheduler-overhead benchmark measures mean state-to-action time of 10.8 μ s for Tabular-Q and 1.567 ms for DQN on observed support. We therefore retain Tabular-Q as a transparent low-overhead baseline, not as a claim of greater algorithmic expressiveness.

Location / evidence

Methods: Policies and Offline Selection; scheduler-overhead overhead table; multi-seed robustness table; Discussion.

Prior reviewer comment

Comment 2: The state representation is not formally defined. For example, it is unclear whether network load, IDS alerts, and queue lengths are jointly encoded, or treated independently, which affects policy optimality.

Author response

The state is now explicit: $s_t = (\tilde{T}_t, h_{t-1}, q_t, \lambda_{t-1}, a_{t-1})$. The first four variables use three bins each and previous action has four values, giving $3^4 \times 4 = 324$ states. Threat is the previous selected detector’s mean attack probability, not ground truth. Current-window labels, phase identifiers, and unselected detector outputs are excluded before action commitment. The physical learned-policy runs reach only four unique state indices because temperature, queue, and fixed scheduled arrival/load remain in one bin each. the reward/state sensitivity analysis additionally removes each state component in turn; all thirty state-ablation policies remain single-action LightLR on held-out support.

Location / evidence

Methods: Decision Information and State Encoding; the reward/state sensitivity analysis state-ablation table; Discussion.

Prior reviewer comment

Comment 3: The reward function is heuristic and lacks theoretical grounding. For instance, combining latency reduction and IDS accuracy without normalization may bias the agent toward one objective.

Author response

The executed reward and every normalization constant are now reported, and we add a pre-deployment pairwise non-degeneracy criterion rather than treating the collapse only as a post-hoc diagnostic. For actions a, b , strict preference reversal on support $\mathcal{S}$ is possible only when the implied exchange threshold $\tau_{ab} = \Delta C_{ab}/w_D$ lies strictly between the minimum and maximum attainable detection-term difference $\Delta D_{ab}(s)$. Under the executed min-max scaling, MedRF requires a 0.7967 balanced-accuracy advantage over LightLR—an almost 80-percentage-point exchange rate determined solely by the declared weights and resource profiles. The criterion can therefore be applied to calibration/validation support before held-out testing rather than inferred from test performance. We also explain why candidate-pool extrema make the exchange rate depend on pool membership and recommend external power/SLO/thermal budgets as normalization references. The reward/state sensitivity analysis retains the empirical sensitivity evidence: 80/85 reward/state policies choose LightLR, while the five detection-only Tabular-Q policies choose MedRF; none replaces the primary policy.

Location / evidence

Methods: Reward and Cost Provenance; Discussion: Reward Specification as a Pre-deployment Check; Supplementary Table S17.

Prior reviewer comment

Comment 4: The evaluation lacks strong baselines. A meaningful comparison should include at least one DRL-based scheduler and one optimization-based method (e.g., MILP or heuristic scheduling).

Author response

The rebuilt shared software evaluation contains four static detectors, CFSM, Tabular-Q, DQN, and a label-informed one-step utility reference that is explicitly nondeployable. The primary physical matrix contains Static-MedRF, CFSM, Tabular-Q, and DQN. Once the learned policies were observed to select LightLR throughout, we added the more relevant missing control as matched-control: a new ten-block matched physical campaign including Static-LightLR. That control changes the interpretation: Tabular-Q versus Static-LightLR has +0.455 J with CI $[-1.861, 2.772]$ J and identical detection, while DQN has +4.525 J $[2.693, 6.357]$ J. We do not call the label-informed reference a full-horizon oracle or claim that matched-control was predeclared before the primary result.

Location / evidence

Methods: Policies and Offline Selection / Post-hoc Strengthening Protocols; matched-control Table and Figure.

Prior reviewer comment

Comment 5: The reported improvements are not statistically validated. For example, results should include mean $\pm$ standard deviation over multiple runs and significance testing (e.g., t-test).

Author response

We separate uncertainty sources instead of calling every seed or window an independent experiment. Classifier training uses one seed. The primary physical campaign uses ten trace-matched blocks and Student- t intervals over the ten block differences; Tabular-Q versus Static-MedRF saves 13.068 J $[9.736, 16.400]$ J and DQN saves 6.227 J $[1.600, 10.853]$ J. matched-control and fixed-rate cascade campaign/variable-load campaign likewise use ten matched physical blocks. variable-load campaign now reports run-level energy SD in the main table because variability is method-dependent. In particular, DQN-minus-Static-LightLR raw energy is positive in all ten variable-load campaign blocks (median +5.72 J; exact sign-flip $p = 0.00195$), but one +392.0-J thermal/power excursion inflates the paired SD to 122.2 J and the Student- t interval to $[-43.2, 131.7]$ J. We therefore retain both the planned t interval and robust direction diagnostic rather than hiding the skew or claiming a precise premium. multi-seed adds ten new policy-training seeds per learned algorithm, but these are software policy robustness rather than physical replications. We do not treat 500 windows or 10,000 microbenchmark timings as independent hardware repeats.

Location / evidence

Evaluation: Statistical Units; primary paired table; matched-control table; Supplementary paired diagnostics.

Prior reviewer comment

Comment 6: Scalability is not demonstrated. The current setup (single device / limited nodes) does not reflect realistic IoT deployments with multiple edge nodes and dynamic traffic patterns.

Author response

We have narrowed rather than inflated this claim. The physical dynamic evidence is explicitly conditional on one Pi 4B 8 GB and TON-IoT feature replay. Repeated blocks on one board do not estimate board-to-board variability or fleet behavior. We no longer use a synthetic fleet analysis to imply physical scale-out validation. Multi-node scalability and cross-device portability are stated as future work outside the evidence boundary.

Location / evidence

Limitations: Physical Measurement and Deployment Scope; Conclusion.

Prior reviewer comment

Comment 7: The convergence behavior of the RL model is not analyzed. Learning curves (reward vs episodes) are needed to show stability and training efficiency.

Author response

All 1,000-episode training curves and validation observations are retained, and the checkpoint rule is explicit. We no longer equate a single smooth curve with algorithm-wide convergence. multi-seed adds ten new seeds per learned algorithm. Selected checkpoint episodes vary across those seeds, while every frozen policy still selects LightLR on the four-state held-out support. This is stronger evidence for realized-policy stability than a single reward curve, but it remains support-limited: global 324-state greedy maps vary, especially for DQN.

Location / evidence

Methods: training/checkpoint selection; Supplement: training figures and multi-seed global maps.

Prior reviewer comment

Comment 8: The integration between IDS and scheduling is not deeply explored. For example, how IDS confidence scores influence scheduling decisions is not formally modeled.

Author response

The scheduler uses the previous selected detector's mean attack probability as lagged threat context. It is combined with smoothed temperature, queue/backlog, lagged arrival/load, and previous action before the next action is committed. This causal ordering avoids using the current selected detector's own output to choose itself. Ground truth is used only during offline training reward calculation and post-hoc evaluation; formal test deployment performs no policy update.

Location / evidence

Methods: Decision Information and State Encoding; Reward and Cost Provenance.

Prior reviewer comment

Comment 9: The computational overhead is not evaluated. In edge environments, latency and resource constraints are critical, yet training/inference costs are not reported.

Author response

We now separate classifier inference from controller decision time. scheduler-overhead times only encoded state-to-action selection. The first scheduler-overhead attempt is transparently retained as invalid because a final validator found a CFSM implementation substitution and missing telemetry. One authorized corrected replacement preserves the original benchmark design and passes 240/240 units; a 5,000-state check reports zero CFSM semantic mismatches. On observed states the mean decision time is $0.739\ \mu\text{s}$ for Static-LightLR, $6.582\ \mu\text{s}$ for CFSM, $10.796\ \mu\text{s}$ for Tabular-Q, and $1.567\ \text{ms}$ for DQN. DQN is much more expensive computationally than Tabular-Q but still uses only 0.157% of the one-second decision interval. We do not convert timing directly into energy. variable-load campaign reinforces this boundary: DQN and Static-LightLR make identical detector choices, but the raw paired energy magnitude is highly skewed by one power/temperature excursion and the idle-adjusted comparison is unresolved; because Controller-only power arm stopped before measurement, no controller-only joule attribution is claimed.

Location / evidence

scheduler-overhead overhead subsection and table; Supplementary Table S15; Limitations.

Reviewer 2

Prior reviewer comment

Comment 1: I'm not convinced the dynamic scheduler actually beats the simplest static option. Looking at Table V, static TinyDT matches the scheduler's F1 on TON-IoT and slightly beats it on CIC-IoT, while using less power and being faster on both. So what is the dynamic scheduler really buying us? The authors frame it as "you don't need to pick a model in advance," but they themselves profiled all four models on the hardware to set up the cost indices—so the profiling effort was already done. I'd like to see a setting where the dynamic approach clearly wins over TinyDT, for example traffic that shifts over time, concept drift, or scenarios where the best static model changes. Otherwise the case for the added complexity is hard to make.

Author response

We agree with the concern and now answer it with a prospectively frozen variable-load campaign rather than a workload chosen after seeing the result. variable-load campaign contains ten paired blocks and six methods; every 500-second run visits $50/100/4000\ \text{rows/s} \times 0.10/0.50/0.90$ attack prevalence exactly once, followed by a fixed recovery cell. A DeadlineGuard rule was frozen from pre-variable-load campaign MedRF latency: use MedRF only when the predicted 100-row-batch service fits the one-second deadline, otherwise LightLR. The result is not favorable to RL. DQN remains LightLR in all 5,000 windows. Tabular-Q reaches new load-related states and selects TinyDT in all 5,000 windows, producing F1 0.7434 and FPR 67.76%. The "fallback" label is now defined precisely: it is not a separate exception branch; unseen zero-initialized Q rows are tied, NumPy argmax selects the first maximum, and TinyDT is the first configured action. In contrast, DeadlineGuard selects MedRF in 3,500 and LightLR in 1,500 windows and records zero deadline misses. Relative to Static-MedRF it uses 43.714 J less per run (95% paired CI 41.418–46.010 J; 2.98%), eliminates the 10.24-pp deadline-miss rate, and loses 1.373 F1 pp. All ten energy differences have the same direction (exact sign-flip $p = 0.00195$). This is now the central positive variable-load campaign comparison. A separate fixed-rate confidence cascade recovers MedRF-like F1 with 1.34% row escalation. Frozen source confirms that it batches deferred rows, but 66.76% of 100-row windows contain at least one deferred row and therefore trigger a near-fixed-cost MedRF sub-batch call; we explicitly scope its energy/latency result to this implementation rather than to confidence cascades generally. We therefore no longer answer the reviewer with "future work": the fixed experiment shows that a simple causal conditional controller can improve the energy/feasibility trade-off against the high-security static baseline, whereas the frozen learned policies do not exploit that opportunity safely.

Location / evidence

Introduction; Evaluation: Physical Campaigns; fixed-rate cascade campaign/variable-load campaign tables; Discussion: Attribution, Adaptation, and Feasibility; Supplementary Tables S19–S22.

Prior reviewer comment

Comment 2: The reward weights are picked without a good explanation, and everything depends on them. The weights $\vec{w} = [0.5, 0.2, 0.2, 0.1]$ are just stated and used throughout. The authors admit in the Limitations that different weights would change the results, but no sensitivity analysis is provided. This is a problem because all the main findings—which model gets picked most often, the power and latency savings, the thermal behavior—flow directly from this choice. Also, the cost indices are normalized so TinyDT = 1.0, which essentially bakes “TinyDT is cheapest” into the reward, and then the paper reports that TinyDT dominates as a finding. Please include a sweep over different weight settings to show the conclusions hold up.

Author response

the reward/state sensitivity analysis provides the requested sensitivity while preserving the original default policy. It contains 55 reward-weight runs and 30 Tabular-Q state-ablation runs. All 85 valid policies are single-action on held-out support. Eighty choose LightLR; the five detection-only Tabular-Q runs choose MedRF. Thus the result does not support weight-invariant adaptive behavior. Instead, it shows that the resource-aware weight families retain LightLR under this support, whereas removing resource penalties changes the selected static detector. We explicitly do not choose a new “best” weight from test results.

Location / evidence

Post-hoc Strengthening Protocols; policy-robustness table; Supplementary Table S17.

Prior reviewer comment

Comment 3: Table III mixes things that shouldn’t be compared directly. The IoT pool models and the UNSW-NB15 reference architectures sit in one table, which makes it tempting to compare F1 across them—but the text itself says you shouldn’t. I’d suggest splitting them into two tables.

Author response

The rebuilt main classification table now contains only the four TON-IoT detector configurations used by the physical scheduling study. CICIoT2023 and N-BaIoT detector/coverage audits remain in the supplement, where their dataset-native dimensions and split boundaries are explicit. We do not compare F1 across an unrelated UNSW-NB15 architecture table or claim a common cross-dataset power model.

Location / evidence

Data Construction and Audit Boundaries; Completed Static Results.

Prior reviewer comment

Comment 4: The CFSM baseline can’t be reproduced from what’s written. The thresholds are described as “fixed” but the actual values aren’t given. Since Figure 3 depends on this baseline, please add the threshold values somewhere.

Author response

The complete rule is now reported: TinyDT initialization; temperature step-down at smoothed 70°C; two complete cooldown windows after a switch; threat threshold 0.7; one-rank up/down movement with boundary holding. Queue/load are logged but not used by CFSM. The unfavorable primary CFSM result is retained without retuning. scheduler-overhead additionally validates the corrected benchmark CFSM against the frozen semantics on 5,000 states with zero mismatches.

Location / evidence

Methods: Policies and Offline Selection; scheduler-overhead audit; reproducibility supplement.

Reviewer 3

Prior reviewer comment

Comment 1: Table 2 shows the values for the Q-learning hyper parameters. The authors should include a discussion justifying the rationale/selection of these values.

Author response

The method now states zero initialization, $\alpha = 0.1$, $\gamma = 0.9$, linear epsilon decay from 1 to 0.01, 1,000 episodes, and the exact validation checkpoint rule. These are baseline choices for the compact discrete controller, not universal optima. the hyperparameter sensitivity analysis adds 35 valid Tabular-Q sensitivity runs over predeclared $\gamma \in \{0, 0.5, 0.9, 0.99\}$ and $\alpha \in \{0.05, 0.10, 0.20\}$ settings. All 35 select LightLR on held-out support; none is used to replace the primary configuration.

Location / evidence

Methods: Policies and Offline Selection; Supplementary Table S18.

Prior reviewer comment

Comment 2: A description/discussion on algorithm 1 seems to be missing from the paper. Including this will help the reader in understanding the proposed IDS scheduling algorithm.

Author response

The executed sequence is now described explicitly: construct the causal pre-action state; commit and log one action; run only the selected resident detector; log prediction, inference timing, and telemetry; update lagged history; and, during offline training only, compute the labelled reward and update the learner. Formal test execution freezes the policy and performs no update. The implementation evidence retains state/action logs and exact source hashes.

Location / evidence

Frozen Detectors and Causal Scheduling; executed-source provenance supplement.

Prior reviewer comment

Comment 3: The rationale for using Q learning to solve the IDS problem may be discussed.

Author response

We clarify that Q-learning does not perform intrusion detection. It selects among four pretrained detector operating points in a compact discrete controller. Its advantages are transparency and low state-to-action cost relative to a neural value approximator. The rebuilt result also shows the limitation of that rationale: under narrow observed state support and the retained scalar utility, the learned selector becomes static. The contribution is consequently the measurement of that boundary, not an assertion that Q-learning is intrinsically preferable.

Location / evidence

Introduction; Methods; Discussion.

Prior reviewer comment

Comment 4: In table V, it is observed that the dynamic algorithm offers reduced power consumption versus static RF. This seems to be counter intuitive. The reasoning for this may be explained by the authors.

Author response

The prior explanation based on frequent runtime mixing is not supported by the rebuilt campaign. The actual primary learned policies run LightLR throughout, whereas Static-MedRF runs MedRF throughout. matched-control verifies this directly: Static-MedRF is +12.471 J [10.611, 14.331] J relative to Static-LightLR, while Tabular-Q is only +0.455 J with an interval crossing zero. The lower primary energy versus RF is therefore primarily detector operating-point selection, not a mechanism that makes MedRF cheaper or evidence of repeated adaptive switching.

Location / evidence

matched-control results; Discussion; Conclusion.

Reproducibility and protocol deviations

The archives preserve all known deviations rather than deleting them. Source-provenance recovery restores 6/6 byte-exact historical executed files. Invalid attempts and authorized replacements in the matched-control, scheduler-overhead, and reward/state studies remain linked, and the multi-seed analysis uses frozen checkpoints without reselection. The reviewer-control archive retains 40 fixed-rate and 60 variable-load valid formal runs plus all attempts. Frozen fixed-rate cascade source also preserves the confidence-cascade implementation: one LightLR batch call plus at most one MedRF deferred-sub-batch call per window, rather than one MedRF call per deferred row. The planned controller-only power arm is explicitly NOT_EXECUTED_PREDECLARED_AMBIGUITY; The TinyDT threshold diagnostic is post-hoc and does not replace threshold 0.5. The reviewer ZIP has 1,912 entries, a 1,911-file manifest, passing CRC/final validation, no policy retraining, and no test-driven selection.

Closing statement

The revised manuscript is intentionally more conservative than the prior submission. Its positive result is the measured feasibility behavior of predeclared conditional controls; its negative result is that the frozen learned policies fail to realize safe adaptation under the same changing workload. Reporting both, together with the Controller-only power arm protocol stop and TinyDT threshold diagnostic calibration diagnostic, provides a more reproducible service-management contribution than the earlier dynamic-switching claim.

Corresponding author: Nuonan Ouyang nuonan.ouyang@my.jcu.edu.au